Why this coordinate

Any autonomous scalar flow $\mathrm{d}\bar{a}/\mathrm{d}\bar{w} = g(\bar{a})$ is straightened by $u = \int \mathrm{d}\bar{a}/g(\bar{a})$. Here $g = (1-\bar{a})^2(1+\delta a)$ gives $u = 1/(1-\bar{a})$ and $\mathrm{d}u/\mathrm{d}\bar{w} = 1 + \delta a$. The height writing's law is NOT autonomous ($\bar{a}'$ depends explicitly on $\bar{w}$), so it has no such coordinate; the state writing's defect and its remedy are two sides of the same property.

Physically $u = (1 + \delta a)(\bar{w} - \bar{w}_0)$ is the gap to the implied wall. The state changes from "how strong is the gain" to "how far am I", and distance propagates by exactly how far one moved.

What it buys

1. The one-signed quadrature ratchet is removed IDENTICALLY, not reduced. $\mathrm{d}u/\mathrm{d}\bar{w}$ is constant in $\bar{w}$, so the one-step integral is exact. Measured on the canonical command trajectory (same Σ, $\bar{a}$-Euler versus exact-u): $+1.641\,\%$ at the end of the descent, $+2.122\,\%$ at the end of the oscillation, strictly one-signed throughout, and exactly first order in Δt ($+1.077/+0.542/+0.272\,\%$ at $\Delta t/2, /4, /8$). Closed form: the $\bar{a}$-Euler step equals $u[k+1] = u[k] + \Sigma + \Sigma^2/u + \mathcal{O}(\Sigma^3)$, and $\Sigma^2/u > 0$ regardless of direction — that is the ratchet. $+2.1\,\%$ is the right size against the 2.8 point gap between arm (a) at $4.68\,\%$ and the true-slope arm at $1.87\,\%$.
2. The predict increment reads $\hat{u}$ nowhere: $\mathrm{d}u = (1 + \delta a)\mathrm{d}\bar{w}$. That is the property the "$\bar{a}'$ from the TRUE height" arm had, obtained here without knowing the true height.
3. u is affine in the unknown wall position, so a Gaussian posterior is closer to the true posterior shape in u than in $\bar{a}$. An EKF in u is a DIFFERENT, non-equivalent estimator — this is the mechanism by which it can win.
4. $(u, \delta a)$ separate cleanly: a wall-position error is a constant offset in u, a δa error is a ramp proportional to accumulated displacement ($F_e(4, 5) = \hat{\Sigma}$).

What it does not buy

No information. $u = (1 + \delta a)(\bar{w} - \bar{w}_0)$ with $\bar{w}$ commanded and free, so $(u, \delta a) \leftrightarrow (\bar{w}_0, \delta a)$ is a known bijection at fixed $\bar{w}$, and Fisher information is invariant under smooth reparametrisation. What changes is conditioning. (That invariance assumes the prior is pushed forward consistently — see the seed section.)

The self-amplification is NOT abolished. The term $-2(1 - \hat{\bar{a}})\Sigma$ is exactly the coordinate's own time-varying Jacobian: a fixed e_u really does correspond to an $e_{\bar{a}}$ that grows $123\times$ between $\bar{h} = 22.2$ and 2.0. What changes is that it moves from the propagation Jacobian to the output map $\partial\bar{a}/\partial u = u^{-2}$, where it does not compound.

The structural limits of the $\bar{a}$ writing survive unchanged: in a hold $\hat{\Sigma} = 0$ so $F_e(4, 5) = 0$ and δa is unobservable through the state path; over a closed oscillation $\sum \hat{\Sigma} = 0$ so the state path carries zero net information about it. Only y_2 sees δa continuously.

Also unchanged: the y_1 gain-correction path through $K_1 = PH_1^\top/S_1$ (measured to supply essentially all of the across-seed spread, and to be correctly priced at 0.04–16 % of the realised optimum); the MA(2) augmentation the controller runs but no derivation describes ($\approx 21\,\%$ of the spread); $\bar{h}_{\mathrm{safe}} = 1.5$ as an underived house value.

Seed, prior, and a well-posedness constraint

δa enters the LAW here, so its prior enters u at $k = 0$ and $P_{u\,\delta a}[0] \neq 0$. With $\hat{u}[0] = 22.22$ and $\sqrt{P_{\delta a}} = 0.5$ the three contributions to $P_{uu}[0]$ are 123.5 (δa), 0.012 (wall position) and $\hat{u}[0]^4 \mathrm{floor}_a^2$; the disturbance term dominates the wall term by 10^4.

floor_a MUST be re-derived for THIS family. The single-curve family of `formC_state_dist` declares 13.05 % RMS / 15.34 % sup; this family, which carries δa, is the one declared in `formC_state_da` at 0.66 % / 4.65 %. The choice matters far more here than in $\bar{a}$ coordinates because the floor is amplified by $\hat{u}[0]^4$:

floor_a	σ_u	$P(u < 1)$	status
0.0306 (envelope sup)	15.11	8.0 %	ill-posed
0.0056 (local at seed)	2.77	≈ 0	well-posed

$u > 1 \Leftrightarrow \bar{a} < 1$, and $u \to 1$ is the singularity of the output map $1 - 1/u$. Eight percent of prior mass at $\bar{a} < 0$ sitting on that singularity is not a width question. In $\bar{a}$ coordinates the same two choices differ only by a factor 5.5 in width; a linear push-forward cannot preserve both tails. The implementation therefore uses the local floor and clamps $u \geq u_{\min}$ (with $\bar{h}_{\mathrm{safe}} = 1.5$ the natural value is $u_{\min} = 1.5$), and clamps $\hat{v}$ likewise.

Reporting $\hat{\bar{a}} = 1 - 1/\hat{u}$ carries a Jensen bias $\approx -P_{uu}/\hat{u}^3$: negligible once converged (0.01 % at $u = 2$, 0.03 % at $u = 4.2$) but 2.18 % at the seed under the sup floor versus 0.07 % under the local floor — the same choice again.

Verification, 2026-08-13

Independent check before implementation. Central claim, S6, S7 and S8 confirmed term by term, including a time-varying similarity transform $F^u[k] = T[k+1] F^{\bar{a}}[k] T[k]^{-1}$ with $T = \hat{u}^2$ that reproduces the printed F_e exactly. Three errors were found and are fixed in the main file:

1. $P_{uu}[0]$ omitted the δa term and $P_{u\,\delta a}[0]$ was set to zero. Inherited from the wrong sibling (see 3), where the zero is correct because δa does not enter the law there. Magnitude-dominant.
2. The delay back-off was labelled exact; it drops the $(\delta\bar{w}_3 - \delta\bar{w}_1)$ channel, which `formB_ws_ref` already declares as a truncation. Measured contribution 0.5–3 % of $\sqrt{R_2}$ (peak 11 %), i.e. ≤ 0.1 % of variance, but it is correlated with the states, so it is bias and not variance. Now declared. Note R_2's $d\,\hat{v}^{-4} Q_{uu}$ exists precisely because H omits that channel, so “exact” and that term cannot both stand.
3. The sibling cross-reference pointed at `formC_state_dist.tex`, whose δa is a different object (additive per step, absent from the law). The $\bar{a}$-coordinate version of THIS law is `formC_state_da.tex`.

Also reported: `formC_state_da.tex` is internally inconsistent on Q_{33} (prefactor $4k_BT/(a_oR)$ in one place, $4k_BTa_o/R$ in its ξ); the main file here is the correct one.

Pre-registration

On the canonical scenario over the 8 house seeds, overall relative gain-error RMS: the same estimator in $\bar{a}$ coordinates gives $4.68 \pm 2.56\,\%$; fed $\bar{a}'$ from the true height it gives $1.87 \pm 0.93\,\%$; production tier-2 gives $2.07 \pm 1.38\,\%$.

lands near 4.7	the gap is information; the coordinate cannot help
lands near 1.9	the gap is conditioning
lands below 1.9	also expected: the true-slope arm still carries the Euler ratchet, which this writing removes as well

Arm (a) against 4.68 % is the clean controlled comparison: same law, same estimator, same measurement chain, differing only in the quadrature and in whether the integrand reads the state.